

Aryan Ashraf

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Technical Skills

Languages: C, C++, SQL, React, .NET, Firestore (NoSQL), Java, Python, Ladder Diagram (PLC), Structured Text (PLC)
Technologies: Git, Docker, Firebase, TensorFlow, Labellmg, Isolation Forest, EPLAN, Arduino, Rockwell Studio 5000, AutoCAD

Experience

Brock Solutions

Apr. 2026 – Present

Software Developer (TypeScript · React · C# · .NET · LLM Agents)

Kitchener, ON

- Built REST APIs with token-based authentication and account-scoped filtering for a customer-facing case management portal.
- Delivered a full-stack support module with a .NET backend and React UI featuring cascading, context-aware dropdowns.
- Wrote an integration test suite covering auth, access-control, and entity validation against the live development environment.
- Built a multi-agent LLM system with an orchestrator-subagent architecture grounded on internal repositories to automate cross-repo analysis and planning.

University of Guelph

Apr. 2025 – Sep. 2025

ML & Assistive Devices Research Assistant (C, C#, SQL, Firestore)

Guelph, Ontario

- Supported a VR mirror therapy app by debugging and extending C# logic in Unity and managing a Firestore backend for patient session and movement data.
- Debugged and maintained Arduino (C) firmware for a smart rehabilitation glove, integrating sensors and real-time control logic validated across 20+ participants.
- Analyzed 100+ temperature sensor readings for a diabetic sock project using SQL and Excel, applying Isolation Forest anomaly detection to identify abnormal patterns.

Projects

University Management System | Java, JavaFX, SQL, Excel

- Designed and developed a role-based university management system with separate admin and user access for managing subjects, courses, students, faculty, and events, collaborating on overall system architecture and implementation
- Implemented authentication, scheduling, and reporting features using Java and JavaFX, integrating SQL databases and Excel-based data import to support efficient data storage, updates, and end-user usability

Real-Time ASL Detection Program | Python, TensorFlow, YOLOv8, Labellmg, Jupyter

- Built a real-time computer vision system for American Sign Language detection using Python on Jupyter, while using TensorFlow, and YOLOv8 & achieving 85%+ accuracy and displaying predictions through a live video interface
- Prepared, annotated, and managed image datasets using Labellmg, creating structured train/test splits and supporting effective model training and validation

Education

University of Guelph

Expected May 2028

Bachelor of Engineering in Computer Engineering (Honours)

Guelph, Ontario

- **Relevant Coursework:** Data Structures and Algorithms (C), Digital Systems Design (FPGA, VHDL, Vivado, Logic Circuits), Programming for Engineers (Java, C++)

Extracurricular

Islamic Relief at UofG

- Supported campus fundraising initiatives by creating marketing content, coordinating charity events, and engaging with students to raise awareness and funds for humanitarian causes

Focus On Engineering Guelph

- Assisted in organizing engineering competitions and contributed to the development and maintenance of a textbook rental website, improving resource accessibility for students